

# Introduction to the GLOBECOM 2023 Optical Networks and Systems Symposium Special Issue

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**This Special Issue contains a collection of invitation-only extensions based on papers presented at the Optical Networks and Systems Symposium at IEEE GLOBECOM held 4–8 December 2023 in Kuala Lumpur, Malaysia. We present a brief introduction followed by an overview of each of the papers.** © 2024 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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The growing demand for higher bandwidth, lower latency, and more robust, cost- and energy-efficient data transmission has fueled the latest innovations in the field of optical networks and systems. This Special Issue features six papers that are extended versions of papers invited from those presented at the Optical Networks and Systems Symposium at GLOBECOM 2023 held in Kuala Lumpur, Malaysia. The papers span a wide range of topics from network planning and recovery to routing and spectrum assignment in various types of optical networks, including optical satellite networks and quantum key distribution networks.

The Special Issue includes a paper by Shrinivas Petale *et al.* entitled “PRODIGY+: a robust progressive upgrade approach for elastic optical networks.” This paper proposes a robust network planning solution to gradually migrate current elastic optical networks operating in the C-band encompassing multi-core fiber-based space division multiplexing and multiband (in particular C+L-band) technologies. The upgrade is split into progressive steps to enable a smooth, non-disruptive migration path while optimizing operational and economic objectives under dynamic growth scenarios.

In another contribution, entitled “Symmetry-free routing and spectrum assignment: a universal algorithm based on first-fit,” George Rouskas investigates the first-fit spectrum assignment problem, first showing its universality. Any spectrum assignment problem variant can then be optimally solved by considering only the connection permutations and applying the first-fit algorithm. This insight forms the basis for a symmetry-free model for spectrum assignment in general network topologies. By eliminating symmetric solutions, the paper introduces a two-parameter routing and spectrum assignment algorithm suited for multi-threaded execution capable of trading off running time and solution quality.

In “Latitude- and time-zone-aware load balancing in optical satellite networks,” Kexin Gao *et al.* investigate the load unbalance issue in optical satellite networks deriving from both varying human activity across different time zones and varying lengths of links connecting satellites on different orbital planes. The latter causes significantly higher utilization of high-latitude inter-satellite links. To solve these issues, a time-zone-based traffic model and a link compensation matrix are introduced, and two novel load-balancing algorithms are proposed adjusting the link weights in optical satellite networks. Results indicate that the blocking probability can be significantly reduced at the expense of a modest increase in latency.

In “Routing problem for reducing significant outages in optical networks and its system analysis,” Katsuaki Higashimori *et al.* introduce a routing method for minimizing the impact of network outages that exceed a given significance threshold. Indeed, while any failure should be avoided, large-scale failure must be particularly suppressed. A routing problem, called least-significant outage routing, is defined. After showing the problem’s NP-hardness, a heuristic routing method is proposed for efficiently solving it, exhibiting a better solution than conventional shortest-path and highest-availability-path approaches.

In “Extension of the  $k$ -SPF algorithm for finding SRLG-disjoint primary and backup route pairs in optical networks,” Hiroshi Matsuura *et al.* propose an algorithm for finding multiple minimum-cost shared risk link group-disjoint primary/backup route pairs between a couple of optical network nodes. Additionally, the algorithm supports the specification of primary/backup transit nodes by a network operator and thus the selection of multiple sections for each primary/backup route. Results indicate that route pairs can be found with

higher probability and that are faster compared with existing solutions.

Finally, in “DRL-based progressive recovery for quantum-key-distribution networks,” Mengyao Li *et al.* address the problem of progressive network recovery in quantum key distribution networks, characterized by peculiar technologies such as optical bypass links and trusted nodes. The combination of these technologies leads to four possible network architectures, for which an integer linear programming model is formulated to optimize the achievable cumulative key rates during recovery. A deep reinforcement learning-based progressive recovery algorithm is also proposed for large-scale topologies, approaching the optimal solution and outperforming baseline algorithms.

All extended manuscripts underwent a separate rigorous peer review process for this Special Issue, and we thank all the reviewers whose meticulous reviews were very helpful in enhancing the submitted articles. We hope you enjoy reading about these latest advances in optical networks and systems that were presented at GLOBECOM 2023.

**Guest Editors:**

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